

TASK 6: Definition of Done — Multi-Language & Multi-Currency Support

Bepha Tools E-Commerce Project

Written by Brian McCarthy

A user story is DONE only when ALL criteria below are satisfied and verified. No user story may be accepted at Sprint Review unless every applicable criterion is met. This DoD covers all 12 user stories in the Sprint Backlog.

SECTION 1: FUNCTIONALITY CRITERIA

Multi-Language Functionality — specific criteria:

1. The target language is fully available as a selectable option in the language switcher on EVERY page of the site.
2. ALL user-facing text is translated — including: product titles, product descriptions, product specifications, navigation menus, category labels, search bar placeholder, filter labels, all checkout page steps (cart review, shipping, payment, confirmation), all form labels, all button text, all error and validation messages, order confirmation page, order confirmation email, shipping update emails, customer support pages, and all FAQ content.
3. Switching to the target language immediately updates all page content without requiring page reload or logout. The language preference is persisted for the session and retained when navigating between pages.
4. All special characters, accented letters, and CJK characters (for Chinese, Japanese, Korean) render correctly with no encoding errors, replacement characters, or broken glyphs.
5. Zero untranslated English strings are visible in the target language — verified by an automated string completeness scan.

Multi-Currency Functionality — specific criteria:

1. The target currency is available as a selectable option in the currency switcher on every product and checkout page.
2. ALL prices are displayed in the target currency with the correct symbol, correct decimal format, and correct thousand-separator convention specific to that currency (e.g., ₩12,500 for KRW with no decimals; ₹1,00,000 for INR using Indian lakh formatting; R\$ 1.250,00 for BRL with comma decimal separator).
3. The complete checkout flow — add to cart → view cart → enter shipping → enter payment → order confirmation — completes successfully end-to-end in the target currency.
4. Exchange rates are sourced from the live API and are no more than 24 hours old. If rates are stale (>24 hrs), a warning is displayed and checkout is blocked until rates refresh.
5. The order confirmation page and email both display the currency code, symbol, amount charged, and exchange rate used at the time of purchase.

SECTION 2: TESTING CRITERIA

Multi-Language Testing — specific criteria:

1. Unit test coverage is 100% for all language-switching components. Tests verify: correct locale string loading, fallback to English for missing strings, no raw translation key IDs visible to users.

2. A native speaker (or certified translation agency) has reviewed and provided written approval for all translated content. The approval document is attached to the user story ticket.
3. An automated string completeness scan has been executed and returns zero (0) missing translation keys for the target locale JSON file. The scan output log is attached to the ticket.
4. Regression tests confirm all existing English, Spanish, and French language content is unaffected by the new language addition.

Multi-Currency Testing — specific criteria:

1. Unit test coverage is 100% for all currency conversion functions. Tests cover: correct symbol display, correct decimal/thousands formatting per locale, correct rounding rules (e.g., KRW rounds to whole numbers, BRL rounds to 2 decimal places), API failure handling.
 2. End-to-end integration tests pass for the complete checkout flow in the target currency: add to cart → checkout → enter test payment → receive order confirmation email in target currency.
 3. Stale-rate scenario test: QA simulates an API failure/outage lasting >24 hours. Verify the system displays a clear warning and blocks checkout until the rate is refreshed.
 4. Regression tests confirm all existing USD, EUR, GBP, and CAD currency checkout flows are unaffected by the new currency additions.
-

SECTION 3: DOCUMENTATION CRITERIA

Multi-Language Documentation — specific criteria:

1. Technical documentation is written and published in the team knowledge base (e.g., Confluence) covering: i18n framework configuration, locale routing setup, translation JSON file location and naming conventions, and a step-by-step guide for how to add a new language in the future.
2. User-facing help center content is available in the target language covering: how to change language settings, FAQ articles relevant to international customers, contact/support instructions.
3. The tech lead has reviewed and approved all technical documentation before Sprint Review.

Multi-Currency Documentation — specific criteria:

1. Technical documentation is written and published covering: currency API provider name and configuration, API key management procedures, exchange rate refresh schedule, rounding rules for each currency, and a step-by-step guide for adding a new currency.
 2. Financial compliance notes are documented covering: which currencies require tax calculation adjustments, PCI-DSS compliance status for each new currency's payment gateway, and any regional financial regulations affecting checkout.
 3. The Finance stakeholder has reviewed and signed off on compliance documentation before the feature is deployed to production.
-

SECTION 4: DEPLOYMENT AND SIGN-OFF CRITERIA

1. Feature flags are configured for each individual language and each individual currency — independently toggleable per environment (dev, staging, production) without a code deployment.
2. All features are deployed to production with zero downtime. A rollback procedure is documented and successfully tested in staging before production release.
3. Monitoring alerts are active for: broken locale routes, stale exchange rates (>24 hrs), and currency API failures. A test alert has been confirmed firing in the monitoring system.
4. The Product Owner has reviewed and verbally approved all completed features in the staging environment. Written sign-off is recorded in Sprint Review meeting notes.